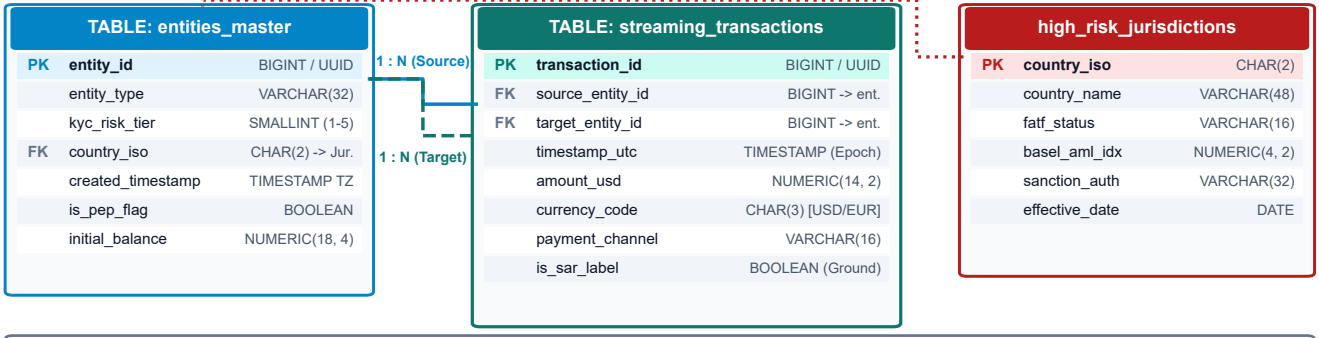
Relational Database Schema (ERD), Causal Time Indexes & 12-D Graph Feature View



DUCKDB / POSTGRESQL TIME-CAUSAL COMPOSITE INDEXES & MATERIALIZED 12-D INVARIANT VIEW

Composite Causal Indexes (Zero Lookahead):

CREATE INDEX idx_source_time
ON streaming_transactions(source_entity_id, timestamp_utc DESC)

CREATE INDEX idx_target_time
ON streaming_transactions(target_entity_id, timestamp_utc DESC)

Execution Guarantee:
O(log |E|) 2-hop causal ego-graph extraction in <0.18 ms

Lookahead Invariant:
Strictly t_e <= t_eval (eliminates future edge leakage)

CREATE VIEW v_canonical_invariants_12d AS (Directed Multigraph):

1. **Flow Regularity:** $\Phi_{\text{flow}} = \ln(1 + \sum A_{\text{out}}/(\sum A_{\text{in}} + \varepsilon))$
2. **Hawkes Arrival Rate:** $\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha \cdot e^{-\beta(t - t_i)}$
3. **Directed Causal Taints:** $s_{\text{fwd}} \in [0, 1]$ **(Dispersion),** $s_{\text{bwd}} \in [0, 1]$ **(Aggregation)**
4. **Graph Multi-Moments:** $[\mu_A, \text{Var}(A), \gamma_A, \kappa_A] \,\|\, [\deg_{\text{in}}, \deg_{\text{out}}, \nu_A, \nu_{\text{io}}] \rightarrow \mathbf{R}^{12}$